Measuring the Production of Scientific Human Capital: New Data, Methods, and Evidence on the U.S. Scientific Training Ecosystem *

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Abstract:

We develop a dissertation-based methodology for measuring PhD populations and use it to present new evidence on who funds STEM PhD training in the United States, how many graduates are trained in areas of strategic national importance, and the effects of public investment in PhD training on PhD production. The U.S. government is by far the largest source of financial and in-kind support for STEM PhD training in America. We identify universities and fields where PhD training has high rates of government, industry, or philanthropic support, and the organizations and universities that fund and train the most PhDs in critical technology areas such as AI, quantum science, and biotechnology. Leveraging variation in government support across agencies and over time, we provide evidence suggesting that increasing government-funded PhD trainees increases PhD production additively or with modest crowd-in. To support further research, we provide public data at multiple levels of aggregation. These data and methods complement existing data collection efforts by national statistics agencies, producing information which is otherwise hard to collect or not systematically observed, and can be extended forward in time and to other countries.

JEL Classification: I23, I28, O31, O32, O33, O38

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1 Introduction

One of the primary goals of international research policy, from its post-World War II origins to the present, is the deepening of countries' research capacity through investments in scientific human capital (e.g., Bush 1945, Steelman 1947, OECD 1963b). The size and shape of the scientific workforce influences not only the pace of scientific advance but also its reduction to practice—particularly in high-tech industries organized around the discovery and commercial application of new science. Reflecting these goals, science and technology (S&T) statistics and cross-national comparisons have long emphasized scientific manpower as a performance indicator (Godin 2002b, 2003). Accordingly, a substantial share of science funding in the U.S. and elsewhere is spent on training new researchers (e.g. National Center for Science and Engineering Statistics 2025b).

The doctoral trainee population has specifically been a subject of analysis since the 1940s, and modern innovation scholarship is increasingly focused on who enters science, to what effect, and how early careers evolve. Much of this research uses either public data, including government surveys and bibliometric datasets, or the UMETRICS sample (Lane et al. 2015). However, what is known is constrained by what is measured. Despite continuous progress—including recent improvements in access to administrative data—many important features of the scientific training ecosystem remain hard to measure. Questions like how many PhDs are trained in frontier or emerging subjects like artificial intelligence (AI) and quantum computing, where they are trained, who funds their training, and who subsequently harnesses this talent are difficult to evaluate using available data and measurement systems. National comparisons over long horizons are also challenged by data limitations. As a result, though the U.S. graduated >37,000 STEM PhDs in 2024 (National Center for Science and Engineering Statistics 2025a), it is difficult to assess whether the U.S. (or other countries) is producing the "right" kind or number of PhDs, what determines the answer to that question, and how policy choices might cause that number to rise or fall.

In this paper, we introduce a new approach to measuring national doctorate production, using the content of PhD dissertations, and apply it to study several features of the U.S. scientific training ecosystem. Dissertation-based measurement presents three advantages: dissertation science provides a marker of the specific expertise a graduate embodies, and a predictor of future scientific research capacity[1]; dissertations are required of all graduates; and recent advances in large language models (LLMs) make their content considerably more accessible. We focus our analysis on U.S. PhD graduates between 1950 and 2022 in the natural sciences and engineering—over one million

[1]Even Nobel Laureates' future prize-winning science can sometimes be traced to their graduate research. For example, Jennifer Doudna (2020 laureate in chemistry, for the discovery of CRISPR gene editing technology) wrote a 1989 PhD dissertation on programmable RNA enzymes for RNA cutting and splicing.

graduates in total. Using data on the near-universe of these graduates, we use our methods to establish new evidence on (i) the set of organizations which finance doctoral research, (ii) the set of graduates producing science related to "critical technologies" considered important to U.S. national defense and economic security, including where they train, and who funds their work, and (iii) the relationship of public investments in PhD training to total U.S. PhD production.

The foundation of our analysis is the ProQuest Dissertations & Theses Global (PQDT) dissertation library, from which we obtain dissertation text and metadata for STEM graduates (i.e., those in the life sciences, physical sciences, mathematical sciences, and engineering).[2] The PQDT sample tracks administrative benchmarks closely in both total graduates and graduates by university and field, suggesting that it approximates the target population.[3] For each of the 1.2 million dissertations in this sample, we observe metadata (e.g., year, subject, school, title) and for a majority we also have access to the full text—which is available for much of the sample pre-2000, and nearly 100% of the sample post-2000. We then develop two LLM-based procedures to measure new features. One assesses each dissertation's association to critical technology areas identified by the White House Office of Science and Technology Policy (OSTP 2024) as U.S. national priorities. The other parses dissertations to identify acknowledgments, extracts the names of organizations which supported the graduate's work (financially or otherwise), and links these entities to the Research Organization Registry (ROR), an independent, consolidated registry of global research organizations. We validate the resulting measures against manual and external benchmarks.

Using these data, we characterize the evolution of these features of the U.S. scientific training ecosystem over the past 75 years. About 50% of STEM graduates currently acknowledge some source of support in their dissertations, the majority (42%) from the U.S. government—a share which is at its highest level since 1980, but significantly below its >50% peak in the 1960s. We view these shares as a lower bound on true (underlying) rates of support, which are potentially underreported in dissertations—a possibility we return to later in this paper. Disaggregating this evidence, we document the largest sources of support in the government, industry, and non-profit sectors. The four largest sponsors of U.S. STEM PhD trainees in our data are the National Science Foundation (NSF, with >90,000 graduates supported since 2000), National Institutes of Health (NIH, 80,000), Department of Defense (DoD, 34,000), and Department of Energy (DOE, 30,000). In all, 12 different government agencies each support more graduates than the largest firm (Intel)

[2]ProQuest until recently had agreements with nearly all U.S. universities to acquire rights to republish their graduates' dissertations, which are now available in ProQuest's single, consolidated repository. In recent years, U.S. universities have begun transitioning to self-publishing, an industry shift which will require a more diffuse effort to collect dissertations from institutional repositories (e.g., to extend our analysis) in the future.

[3]PQDT graduate counts match the SED through 2005, and continue to add to roughly 90% of SED totals in the 2010s, with the gap explained by the attrition noted in the previous footnote.

and the largest non-profit (HHMI)—reinforcing that U.S. scientific training, and in turn the future scientific workforce, is heavily underwritten by the U.S. government.

Continuing this evidence, we document universities and subjects with higher and lower rates of government and industry support. The relative balance further indicates which universities and fields are more exposed to changes in funding from each sector—with subjects such as astronomy and astrophysics being particularly government-reliant, pharmaceutical sciences and automotive engineering being relatively more industry-supported, and subjects such as geology and materials science being supported by both. In nearly all subjects—and at all U.S. universities—government support outpaces private sector support, but the rates of government and private sector support correlate with each other, likely reflecting either subjects' scientific potential or resource requirements, but potentially also indicating correlation in demand across sectors.

Using our classification of graduates to critical and emerging technology areas, we identify the top training grounds and sources of support in AI, quantum science, and other areas since 2000. We measure the size of the U.S. PhD pipeline in these areas and highlight institutions and organizations which are ostensibly important to sustaining U.S. leadership in critical technology-related science. In nearly every critical technology area, NSF is the largest recognized source of support, and the top five sources are nearly always government agencies. The institutions which rank highest in their production of critical technology-related PhD graduates are MIT, Stanford, and UC Berkeley, though many other institutions rank highly in specific specialties.

Leveraging variation in individual government agencies' funding priorities (e.g., NIH primarily funds biomedical and health science, and DoD engineering) and each agency's total number of graduates supported each year (a reflection of budgets), we estimate the effect of government support for PhD training on total PhD production, using a shift-share design. The elasticities we estimate imply that across U.S. higher education, total PhD production scales more than proportionally with government-supported graduates in each cohort. If we account for potential underreporting of support in our dissertation sample, the evidence suggests crowd-in of an additional 0.2 unfunded graduates per government-funded graduate—a pattern which is broadly consistent with what prior research has found for specific programs and settings (e.g., Blume-Kohout and Adhikari 2016).[4] This evidence adds an additional datapoint to the wider literature estimating the impacts of public

[4]This result is broadly consistent with the results of Blume-Kohout and Adhikari (2016), who find a roughly 1-for-1 effect of increasing NIH fellowships and traineeships on total PhD enrollment, and with other work relating public funding for PhD training to enrollment (e.g., Freeman et al. 2009) and career outcomes (e.g., Graddy-Reed et al. 2021, Kim et al. 2022). We expand on this work with population-level measures of funding sources and aggregate PhD production across all fields and over longer horizons—providing a view of the overall system—but with the limitation that we cannot reliably measure specific funding mechanisms (e.g., fellowships vs. assistantships), whose impacts can vary (Blume-Kohout and Adhikari 2016, Kim et al. 2022).

R&D, in this case reinforcing that public investment in scientific training is one of the main factors shaping the size and composition of the future scientific workforce.

In introducing new measures of U.S. PhD production, this paper adds to a long history of advances in S&T statistics, where the measurement of scientific manpower has been a long-running priority (e.g., see National Science Foundation 1959, Freeman 1962, Freeman and Young 1959 for early discussions, and Godin 2002b, 2003 for a more extended history). As Christopher Freeman (founder of this journal, one of the major early influences on the development of national S&T statistics, and author of the original Frascati Manual; see OECD 1963a and Godin 2008) has written, "The first elementary step towards improving the rationality of this process [science policy], towards making these choices more conscious and more carefully considered, is the systematic collection of statistics on the deployment of scientific manpower, and on the expenditures on different branches of scientific activity," further noting that "These statistics must be collected in a great variety of breakdowns" (Freeman 1968, as quoted in Godin 2002a, pp. 12-13).

Although the NSF administers large-scale surveys towards this end like the Survey of Earned Doctorates (SED, an annual approximate-census of graduating PhDs) and the Survey of Doctorate Recipients (SDR, a periodic/biennial survey which samples past graduates), surveys are expensive to administer, have gaps in what they measure, and are difficult to link to other data sources. UMETRICS has more detailed measures of how research is financed, but is limited to universities which have opted into reporting, reports financial transactions only (not in-kind and other support), and suppresses personally-identifying information (like names) for privacy reasons, making linkages to external datasets challenging.[5] Against these alternatives, dissertation-based data have multiple strengths (and some weaknesses, which we will discuss momentarily). Outside of the U.S., large surveys and UMETRICS-like projects may also be too expensive or practically infeasible in many countries. Dissertation samples provide an alternative, complementary lens into doctoral training and the early career scientific workforce which can be applied to any national education system—as other scholars have recently begun to do (e.g., Martínez et al. 2025).

With respect to measurement, this paper explores the capabilities that recent advances in LLM-enabled natural language processing offer for extracting information from large, unstructured texts (like dissertations) to identify target features. This can be done for many more features than we considered in this paper, as well as a wider range of scientific texts.[6] With this paper, we are also publishing new data with information on PhD production by university, field, and university-field

[5]Although UMETRICS has been linked to PQDT, it omits PQDT identifiers to preserve anonymity, such that we were unable to link our measures to the UMETRICS data for comparison.

[6]Research is already moving in this direction, given new opportunities to evaluate publications (Dagdelen et al. 2024), lab notes (Jalali et al. 2024), and even LaTeX markup (Yang et al. 2025).

over time, including measures of (i) graduates associated with each of the critical technology areas specified by the OSTP (2024) and (ii) graduates supported by specific federal agencies and/or with industry or philanthropic support. Beyond this paper, our methods and data are also extensible to other universities and national education systems, to other degree levels, and to future graduates: the main input required is a sample of theses or dissertations.

There are nevertheless limitations to this approach. One is that it requires complete or representative samples of written work from the population studied. Though PQDT appears to contain the near-universe of U.S. STEM PhD graduates through the mid-2010s, a growing number of universities since then have stopped automatically submitting dissertations to ProQuest and instead publish them in open access university repositories, in response to federal guidance (see OSTP 2013, 2022) and the broader open science movement. This attrition is modest in our sample period, and not systematically related to research funding, making it unlikely to affect our findings beyond a (modest) underaccounting. However, filling these gaps and extending analysis of the U.S. doctoral training system forward will likely require expanding data sources to more of these repositories—a possibility we are exploring in ongoing work.[7] In our particular application of identifying research funders, a second limitation of our approach is that similar to survey data, it measures only the sources of support that graduates self-reported—and our evidence suggests there is some under-reporting. As in similarly-impacted survey settings, however, this limitation can be accommodated with some reweighting on reporting rates, an approach we explore in our analysis.

Notwithstanding these limitations, we believe this paper makes a useful advance in both methods and findings. Despite long-running interest in critical technology assessment across many corners of government (Fuchs 2022), systematic measurement techniques are still developing. The methods we develop can be adapted and applied in other settings. The evidence may also have applied value: evidence on the state and organization of critical technology scientific training in the U.S. can inform academic and public understanding of U.S. science, and potentially influence research funding from all sectors by identifying pockets of strength and gaps in capability. Evidence on sources of funding offer a reference point for understanding the potential impacts of large changes in public funding for doctoral training—and at the time of writing (June 2026), large cuts to U.S. graduate student funding are already in progress (Mervis 2025, Bhatia et al. 2025).

We proceed as follows. Section 2 connects our work to previous research on doctoral training and early career scientists. Section 3 describes our data sources, describes measurement methodologies,

[7] The contents of this paper point to the value of national dissertation repositories, which some concurrent research is beginning to use—especially research under the European Doc-Track program (see Martínez et al. 2025 for a project report, and Corsini et al. 2025 for an application). National dissertation repositories are more common in countries with national university systems, though the effect could be replicated in the U.S. with a national depository program similar to the Library of Congress' mandatory deposit requirements.

summarizes the performance of our methods in validation tests, and discusses limitations. Section 4 uses these measures to introduce new facts about U.S. STEM PhD production, examining the level and composition of new trainees and the impacts of public investment in scientific training on PhD production. Section 5 describes the data we are releasing with this paper. In Section 6, we briefly review the paper and then suggest directions for future research.

2 Prior Research on the Scientific Training Ecosystem

Scientific manpower has been a central concern of innovation policy since World War II, which has particularly emphasized the importance of public investment in scientific human capital to advancing national defense, economic, and public health objectives (e.g., Bush 1945, Atkinson 1990). Doctoral training has accordingly been a focus of policy studies throughout this period, including a large and growing academic literature. Economically, knowledge workers in STEM fields are broadly recognized as an engine of innovation (e.g., Bianchi and Giorcelli 2020, Lubczyk and Moser 2025), and their finite population a limiting factor on economic growth (Romer 1990, Jones 2009, Akcigit et al. 2025). Consistent with this view, prior research has empirically linked the size of the STEM workforce to broad-based productivity gains, profits, wages, entrepreneurship, and other economic improvements (Winters 2014, Peri et al. 2015, Balsmeier et al. 2025).

Identifying the forces that shape the STEM scientific workforce is thus a first-order interest of both research and policy. A large share of work on this question has gravitated to immigration policy, on the observation that immigrants comprise a large, disproportionate, and disproportionately impactful share of U.S. scientists (e.g., Stephan and Levin 2001), inventors (e.g., Hunt and Gauthier-Loiselle 2010, Kerr and Lincoln 2010, Bernstein et al. 2022), and entrepreneurs (e.g., Azoulay et al. 2022). In the realm of science, prior research has examined not only immigrant scientist productivity (e.g., Kahn and MacGarvie 2016, Agarwal et al. 2023) and spillovers (e.g., Ganguli 2015), but also the arguably more elementary question of which foreign students come to the U.S. for graduate studies, who stays, and why (e.g., Gaulé 2014, Ganguli and Gaulé 2019, Bostwick et al. 2024). A common theme in this literature is that immigration policy is a key determinant of the foreign-born STEM PhD population (Kahn and MacGarvie 2020). Alongside the aforementioned evidence on the impacts of foreign-born scientists, this recognition has driven recurring calls for visa reform as one way to grow the PhD population (Glennon 2024, Nice 2025).

Despite the emphasis on immigrants, a majority of U.S.-trained STEM PhDs are U.S. born. An alternative path to growing the STEM workforce is to encourage homegrown entrants, and a separate body of research has examined the effectiveness of policies supporting entry into STEM fields, from childhood education (Dynarski et al. 2013) to curricular reforms (Goodman 2019). One of the most direct levers for shaping the scientific workforce is public funding (Bush 1945, Steelman 1947). As Myers (2020) observes, funding can shift the direction of scientists' work. Though existing researchers' switching costs are large (Goolsbee 1998, Myers 2020), trainees and potential trainees may be more responsive—especially if younger workers are flexible and adaptive to changes in demand, as is often the case in other sectors. Consistent with this idea, Dugoua et al. (2025) have found that windfalls to the Department of Energy's R&D budget have historically increased PhD production in DOE-related fields; Cook et al. (2026) relatedly show that university-fields with more federal research grants in a given year expand their capacity and graduate more PhD, MA, and BA students in subsequent years. More generous funding environments in graduate school may also affect students' career progression (Graddy-Reed et al. 2021).

Reflecting long-running recognition that science is important to national outcomes, the NSF has been tasked with measuring it since the agency's inception. The NSF-administered Survey of Earned Doctorates specifically focuses on measuring U.S. PhD production. Despite its long history, many features of the U.S. scientific training system are not systematically known or measured—in part due to the intrinsic limitations of surveys, including their length and complexity. Key questions—such as (i) how many graduates does the country train in frontier, nationally-important areas; (ii) where does this training occur; (iii) who funds this training; and (iv) how many trainees leave the U.S., and with what consequences—remain difficult to systematically answer with existing data. Where some of these gaps have been partially filled (e.g., Chang et al. 2019 present statistics on the funding sources of PhD graduates at seven midwestern U.S. universities in 2011-2012, using UMETRICS data, as a demonstration case), data limitations continue to make it difficult to produce comprehensive national accountings, and the limited opportunities to systematically measure graduates' science have similarly made it difficult to evaluate its nature or impacts.

These are among the gaps we aim to address in this study and a concurrent study of postgraduate migration by U.S.-trained PhDs (Shvadron et al. 2025b). We also introduce new methods and data which others can use to continue exploring these and other themes. Related research loci which we do not address here include scientist career choice (e.g., Roach and Sauermann 2010, 2017, Sauermann and Roach 2012, 2016), how training environments affect career outcomes (e.g., Gaule and Piacentini 2018, Fry and Glennon 2025), and the degree of and consequences of diversity in STEM scientific career pipelines (e.g., Ceci et al. 2014, Buffington et al. 2016, Kahn and Ginther 2017, Bostwick and Weinberg 2022, Stansbury and Rodriguez 2024). The data we are releasing with this paper may be useful for exploring these questions with respect to emerging science-based technologies and the research funding ecosystem.

3 Measuring the Scientific Training Ecosystem

Our first goal in this paper is to build a near-population sample of U.S. STEM (Science, Technology, Engineering, and Mathematics) PhD dissertations since the mid-20th century. Using this sample, we then aim to measure characteristics of PhD graduates which are not elsewhere observed or are hard to observe at the same scale, and for which comprehensive statistics therefore cannot otherwise be produced. We accordingly aim to do so in systematic ways which can span universities, fields, funders, and time.[8] Our emphasis is mostly on two features and phenomena: (i) measures of graduates' financial sponsors, and (ii) measures associating graduates' dissertation science to technology areas which according to the White House Office of Science and Technology Policy (OSTP) are important to defense and/or economic security.[9]

To produce this sample and these measures, we combine data from three sources: ProQuest (a commercial provider of global dissertation data), OpenAlex (an open-access bibliometric database; Priem et al. 2022), and universities' own institutional repositories. From these sources we compile a nearly-complete sample of U.S. STEM PhD graduates over the past 75 years, including dissertation text for many of these graduates pre-2000 and nearly all post-2000. To evaluate sample completeness and some dissertation-derived measures we additionally collect more aggregated data from the Survey of Earned Doctorates (SED) and Survey of Graduate Students and Postdoctorates in Science and Engineering (GSS) as well as individual-level data from one large, nationwide fellowship program, the NSF's Graduate Research Fellowship Program (GRFP).

3.1 Base sample: PQDT

Our main data source is ProQuest Dissertations & Theses Global (PQDT). For nearly a century, ProQuest has been acquiring and re-publishing dissertations of U.S. PhD graduates via licensing agreements with universities. The ProQuest dissertation database is the largest collection of digitally catalogued PhD dissertations in the world, and it includes extensive metadata on each dissertation

[8] In addition to complementing existing data sources on U.S. PhD graduates—including administrative data, such as NCSES surveys—these data sources and methods can likewise substitute for some administrative measures in other countries or contexts where data collection infrastructure is costly and less developed.

[9] The term "critical technology" is indicative of technology that a policymaking organization considers important to national or economic security. The terminology traces to the U.S. Export Administration Act of 1979, which required the Department of Defense (DoD) to provide a list of "militarily critical technologies" to the Department of Commerce to ensure they are export-restricted. This requirement was eliminated by later legislation, and the terminology has morphed to other applications. The OSTP list we use was developed in 2024 by the U.S. National Science and Technology Council, a committee which coordinates science and technology policy across the executive branch. In parallel, DoD's Office of Strategic Capital has also published a list of critical technology areas it considers important to national security, many of which overlap with the OSTP list. The CHIPS and Science Act of 2022 enacted by Congress similarly identifies key technology focus areas, which also intersect. One advantage of the OSTP guidance is that it represents cross-governmental policy. A second is that it includes more detailed descriptions which are useful for linking PhD dissertations (or other textual inputs) to technology areas.

(e.g., author, institution, degree earned, title, abstract, subject). We identify 1.2 million STEM PhD dissertations in PQDT between 1950 and 2022 using degree letters, which we manually filter to PhD and PhD-equivalents, and graduates' self-reported subjects, which we manually crosswalk to SED major fields and filter to natural sciences and engineering.[10] Using IPEDS data, we further restrict to PhDs issued by institutions which ever held an R1 or R2 Carnegie classification post-2000, as well as PhDs from doctorate-granting medical and engineering schools.[11]

By comparison, the SED reports roughly 1.3 million STEM PhD graduates over this period.[12] Relative to the SED (and its underlying, de-identified microdata), a dissertation sample like PQDT presents two distinctive opportunities for analyzing doctoral training systems. First, we observe the dissertation science itself. Second, it provides enough identifying information (most importantly, names) to link graduates to their postgraduate careers. This enables us to do so for more graduates than other NSF instruments such as the SDR, and in more ways.

The SED nevertheless provides a useful benchmark for evaluating the completeness of the PQDT sample. Figure 1 compares annual PQDT STEM dissertation counts to SED totals (defining STEM as above; see Appendix A for a more detailed accounting of the subsumed fields). The PQDT sample tracks the SED population closely until roughly 2010, after which a modest gap emerges—though PQDT still accounts for 90% of the SED population through 2022.[13] Appendix Figure A.2 disaggregates this comparison to individual fields and finds similar patterns at the field level. Together, these comparisons suggest our PQDT-based sample approximates the underlying population, notwithstanding modest undermeasurement since 2010.

[Figure 1 about here]

[10]To link subjects to fields, we use the SED's own crosswalk between research subjects and major fields to the extent possible—which covers most of the PQDT sample—and judgment otherwise.

[11]These institutions together account for over 99.5% of the unrestricted sample.

[12]The SED is an annual survey of graduating PhD students at U.S. universities, and has been administered to the population of graduates in all fields (nearly 60,000 in 2022) since 1958. In practice it is a near-census of U.S. PhD graduates, with a >95% response rate through the mid-1990s and 90% today.

[13]Closer examination of the PQDT sample suggests this difference is mostly attributable to a small but growing number of universities ending their relationship with PQDT, such that their graduates' data are no longer automatically included. This change in turn reflects a movement of universities towards making research freely available, typically on university repositories in addition to or in place of commercial publishers. Much of this movement is in response to either state law or to federal guidance from OSTP in 2013 requiring they do so for federally-funded research within one year of publication (OSTP 2013). Updated guidance in 2022 eliminated this embargo (OSTP 2022). PQDT sample completeness may thus decline going forward. In ongoing work, we are developing a data collection and processing pipeline to collect dissertations from university repositories, one at a time, initially to backfill gaps in our PQDT sample since ca. 2010—especially with respect to a handful of important universities which exit the PQDT sample early, such as Georgia Tech (which does so in 2013)—and eventually with the goal of developing a sustainable data collection program that can substitute for PQDT in scholarly research.

3.2 Dissertation text

Existing understanding of scientific training in the U.S. innovation system is largely shaped by the available data. Surveys like the SED are nearly complete in their sampling and collect a rich set of information on graduates' background and immediate future plans, but offer limited insight into their science. PQDT's tabular data provides basic identifying information about doctoral graduates (e.g., name, institution, subject, graduation year) which can be used in more flexible ways (e.g., linking to other data sources) and is increasingly being used in research, but these data are likewise limited in what they measure. To our knowledge, no prior work has made systematic use of dissertation text, which—as we will show—presents opportunities to measure features of doctoral research, training, and trainees that are not elsewhere observable.

An important feature of PQDT for our purposes is thus that in addition to metadata, it also provides dissertations' full text. An added benefit of ProQuest having been the main provider of dissertation publishing services over the last century is that dissertations from hundreds of different universities generally share a common document structure, which enforces some consistency in reported information beyond science as well. For example, most dissertations have an "Acknowledgments" section (among others), where authors thank funders and other supporters of their research and training. These acknowledgments are a crucial resource for this paper, as they present an opportunity to systematically measure PhD graduates' sources of support.[14]

Dissertation text (beyond metadata) is available for most, but not all, of our PQDT sample: of the 1.17 million dissertations in our 1950-2022 sample, we have the text of 870,000 (75%) in total, and roughly 96% since 2000, which will be a focal period for much of our analysis. Where we use pre-2000 data, gaps are unlikely to be a problem for our analysis provided there are no systematic differences in the set for which text is available: as long as sampling is as good as random, the data can be used to produce representative estimates. Even with systematic differences in data availability, sample weights (e.g., inverse propensity score weights) can be constructed and used to rebalance the full text sample to resemble the population (on observables).

An important question is then what explains why some dissertations have full text, and others do not. Guidance from the data provider and evidence from the data vary a bit with respect to this question. According to ProQuest, this variation is a result of (i) differences in indexing methods—with some dissertations historically having been stored by ProQuest with full text, some with abstracts, and some with identifying data only—and (ii) idiosyncratic variation in what dissertations

[14]Though nearly all dissertations include acknowledgments, there are nevertheless exceptions. When an acknowledgments section is missing, funders are often mentioned in the dissertations' biography section in footnotes to the text—cases which we can also identify by processing dissertations' complete text.

have been retrieved from microfilm. Figure 2(A) shows the share of PQDT dissertations over time in each indexing category, as indicated by ProQuest's dissertation-level identifiers (see Appendix A for details), which suggests that dissertation text may be available for 90%+ of its sample since 1965. In practice, however, we do not reach this level of completeness until ca. 2000. Figure 2(B) shows full text availability in the data, with the share of graduates each year for whom we have dissertation text, by field. Full text is available for around 40% of our sample until 1990, when it discretely jumps to 60-80%, and 1996, when it jumps to ≈100%.

[Figure 2 about here]

We explore one additional effort to grow the full text sample: for seven universities with the most PQDT graduates without dissertation text (MIT, UC Berkeley, UCLA, Cornell, U. of Minnesota, Michigan State U., and U. of Maryland), we visited their online repositories to try to retrieve it. Though Figure 2 indicates that most of these gaps are pre-2000, one exception was MIT, an important institution whose graduates were missing dissertation text throughout our sample. Using this approach we backfilled a large number of dissertation texts for MIT and Michigan State graduates but saw more limited success with other universities (Appendix Table A.1). We incorporate these results into our sample, but given the high cost of one-by-one, university-specific full text retrieval and mostly meager results—and that this mainly fills gaps in the pre-2000 era—we have not expanded this effort to the full sample at this time.

3.3 Linking PhD graduates to technologies

Using the data inputs described above, we aim to produce two new measures. One is to associate PhD graduates to critical technology areas, based on the relationship of their dissertation science to the above-noted 18 technology areas identified by OSTP (2024) as militarily and/or economically important. Similar to Aiken et al. (2024), we develop an unsupervised LLM-based classifier which (i) summarizes dissertations, (ii) uses this summary to evaluate whether each dissertation is related to each of the OSTP technology areas, and (iii) filters these candidate associations based on detailed subfields within these technology areas which the OSTP guidance describes.

Roughly 43% of dissertations in our sample post-2000 are linked to OSTP critical technology areas by this method. The largest is *Biotechnology*, the fastest-growing is *Artificial Intelligence* (AI), and other large categories include *Advanced Materials* and *Semiconductors and Microelectronics* (see Appendix Figure A.6 for a full list). Though critical technology classification lacks a ground truth to test against—"critical technologies" do not have an obvious or naturally occurring taxonomy, nor a clear rubric for evaluation—the method nevertheless validates well against both intuition and independent external measures. The technology linkages our procedure produces are intuitive when read side-by-side with the dissertation titles, abstracts, and text. Using OpenAlex data, we also show in Appendix A that PhD graduates we associate to particular areas are systematically concentrated in universities with the most publications in those areas.[15]

This approach is one among several methods of linking science or scientists to specific categories of technology. A common alternative is patent citations (Marx and Fuegi 2020, 2022), which can be flexibly used with patent classifications as a measure of technology space, though it is limited to science that is cited by patents. More similar to our method, Fry and Glennon (2025) study a sample of PhD graduates in AI based on keyword searches of PQDT dissertations from the 20 highest-ranked U.S. universities in AI-related fields. Our LLM-based classifier essentially systematizes this approach, and may be particularly useful for multidimensional classifications of science (e.g., linking science to many technologies), for classifying large corpora, and for classifying scientific outputs that do not get cited by patents—such as dissertations.

3.4 Measuring dissertation sponsors

The second feature we aim to measure is PhD graduates' research sponsors, including both financial and in-kind support (examples of which might include research inputs, equipment usage, API credits, etc.). Our motivation for doing so is that existing data sources are limited in completeness or resolution. Though the SED includes questions on how respondents financed their doctoral education, the survey has not provided response options to indicate specific funders since 1997. Even then, response options were limited to only a few federal agencies—and as we will show, doctoral training is supported by a much wider range of organizations. Another resource for measuring graduates' sources of support is UMETRICS (Lane et al. 2015), which provides administrative data on 45 universities' financial transactions, including funding from specific sources and transfers to specific employees. UMETRICS provides higher resolution information than SED but for a smaller number of universities and years, and with a changing sample as universities join or leave the program (as of April 2024, 27 universities were actively contributing data). Data privacy considerations also limit the degree to which the data can be linked to external sources. As a result, despite its specificity, UMETRICS is also limited in its suitability for making broad assessments of the U.S. higher education system and evaluations over long horizons.

Dissertation acknowledgments present a potential opportunity to do so over a larger sample of graduates and for a wider range of organizations than existing data sources permit. More complete data on how doctoral training is funded is of both research and policy interest (e.g., Blume-Kohout

[15] We demonstrate this for quantum science in Appendix Figure A.8, plotting universities' number of quantum science PhD graduates against quantum science publications—whose correlation is 0.9.

and Adhikari 2016). For this reason, we develop a methodology to systematically extract this information from dissertations. This approach may also have drawbacks relative to administrative data such as UMETRICS, including incomplete reporting rates, which we will explore below. On net, however, the evidence suggests that despite some imperfections, the data and methodology we develop towards this end yields a broader, more detailed, and more expansive record of how doctoral education is financed in the U.S. than is currently otherwise available.

Many readers of this paper may recognize dissertation acknowledgments from their own thesis. To distill the basic idea behind our measurement, Figure 3 provides an example acknowledgment from the dissertation of Ian Buck, a 2005 Stanford University computer science graduate (now a senior executive at NVIDIA) who in his PhD created the precursor to NVIDIA's CUDA platform, a parallel computing architecture that allows GPUs to be used for general computing and is essential to modern AI. Figure 3 identifies several organizations which funded Buck's work, including DARPA, the Department of Energy (DOE), the Advanced Simulation and Computing program (ASC, a supercomputing facility that is also under DOE), NVIDIA, and ATI Technologies (a former semiconductor company specializing in GPUs, later acquired by AMD). Our task is to identify these funders, consolidate them to their ultimate parents (e.g., grouping ASC up into DOE), and classify them into sectors (government, industry, and non-profits/foundations).

[Figure 3 about here]

To flexibly identify research sponsors across our dissertation corpus, we develop a pipeline that leverages natural language processing tools and large language models (LLMs).[16] We process dissertations one at a time, working initially with the entire text. Our procedure for each dissertation takes the following steps. First, we partition the dissertation into sentences and identify "acknowledgment sentences". To isolate potential acknowledgment sentences, we applied a rule-based string matching approach by splitting the text into sentences (using NLTK's Punkt Sentence Tokenizer; Bird et al. 2009) and identifying sentences with tokens such as "fund", "grant", "thank", and so on. This procedure tags roughly 103 million potential acknowledgment sentences.

We next identify named entities within these sentences, classify entities into sectors, and identify the type of support and grant identifiers (where provided), using the recent open-source LLMs Solar 10.7B and Smaug 34B (Kim et al. 2023, Pal et al. 2024). We established a procedure that uses the LLM to verify that a sentence acknowledges support from an external entity, identify the supporting entities, categorize them by organization and support type, and record grant and contract identifiers

[16] A more detailed description of our methodology, including the LLM prompts we use, and our approach to consolidation and classification, is provided in Appendix A. Here we present an abridged summary.

where applicable. We used the VLLM Python package (Kwon et al. 2023) and the LM format enforcer (Gat 2023) for efficient batch inference and to ensure the model produces a well-structured JSON schema. The LLM output indicates that 11 million of our potential acknowledgment sentences (12%) indeed acknowledged support or were part of a biography section. Among all sentences, the LLM extracted 9.3 million mentions of organizations (4 million unique), with 4.8 million of these mentions originating from sentences classified as support or biography.

We then consolidate named entities (accounting for name and spelling variants by grouping them together) and link them to external firm registries. We consolidate extracted entities using the Smaug 34B model and match them with the Research Organization Registry (ROR) and Wikidata to disambiguate institution names and obtain external identifiers.

3.4.1 Example application

Table 1 demonstrates the performance of our procedure on the acknowledgment text in Figure 3. Each row is an extracted funder, and the columns report the name, sector, type of support, and the associated ROR organization (where matched to ROR; otherwise blank). Comparing the text to the first three columns of this table, we find that our procedure correctly extracts nearly all entities by name, and those it identifies are classified into the correct sectors and type of support. The one exception is ASC, which our procedure has difficulty classifying due to ambiguity of the acronym and its relative infrequency as a research funding organization—one which we ourselves were only able to define with further research and contextual understanding.

[Table 1 about here]

3.4.2 Validation

We validate this approach in multiple ways. The first is by comparing extracted entities to manual classification. In Appendix A.5.1 we identify funders by hand for a random sample of 500 dissertations and compare the share of this sample with government, non-profit, and industry support according to our manual and automated measures. The results show substantial similarity, with 31-32% of this sample classified as reporting government support under both methods, 11-12% non-profit support, and 8-9% industry support—and further agreement at the entity level—suggesting the LLM classifier can reliably approximate manual extraction.

We undertake two further validation exercises. In Appendix A.5.2, we collect data on all NSF Graduate Research Fellowships (GRFs) awarded since the beginning of the program, link awardees to PQDT, and evaluate how many of these graduates acknowledge NSF support in their dissertations.

Based on a simple textual search, we find that among the matched NSF graduate fellows, 60.8% mention the NSF in their dissertation. Comparing these mentions to our LLM-based extraction, we find that our procedure identified and linked 98.6% of these mentions.[17] These results provide further validation for our approach while also highlighting some of the limitations we face, particularly with respect to underreporting—an issue we will return to momentarily.

Finally, in Appendix A.5.3, we compare our sample to the NSF's Survey of Graduate Students and Postdoctorates in Science and Engineering (GSS). Under the GSS, university administrations annually report to NCSES the number of enrolled graduate (master's and doctoral) students in each major field with different sources of funding, including broad categories like institutional support, student loans, or self-support and particular federal agencies like HHS and NSF. Though the GSS sample (which consists of all enrolled students rather than only graduates, and includes master's students) is broader than the sample we collect in this paper, we can nevertheless test whether the data are broadly consistent across the two sources. We correlate across the two sources the share of federally-supported students with support from specific agencies (e.g., DoD, DOE, HHS, NASA, NSF, USDA), at the university-field-year level, between 2000 and 2022. Appendix Table A.11 shows that these sources correlate strongly, with estimates indicating one-for-one correlations for HHS and NSF funding, and near one-for-one for other federal agencies.

3.4.3 Limitations and opportunities for improvement

Collectively, this evidence suggests to us that our procedure can extract acknowledged research sponsors with a high degree of precision and completeness.[18] The resulting data nevertheless have limitations. One of these limitations is a result of our reliance on PQDT: as previously noted, despite being the most comprehensive source of information on U.S. PhD graduates over the last century, the set of universities submitting dissertations to PQDT appears to recently have begun shrinking. In ongoing work we are exploring ways to address this gap, including by collecting dissertations and metadata from individual universities' institutional repositories.

A second limitation more endemic to our approach is the risk of underreporting: graduates may forget or otherwise neglect to acknowledge all sources of support in their dissertations (much as graduates may do the same in their SED responses, or journal publication authors might in research articles, etc.). The comparison of our measures to official NSF GRFP award lists (Appendix A.5.2) suggests such underreporting can potentially even be substantial.

There are two strategies we consider to address underreporting. One is to simply report the data

[17] The remaining 1.4% of mentions that were not picked up by our LLM-based procedure were due to unconventional wording in acknowledgments or mentions of the NSF without an indication of funding.

[18] Note that this process can also be applied to other scientific corpora as well.

as they are—an approach which will generally yield conservative measures (i.e., underestimates) of the share of graduates supported by any given agency or organization. The second is to use our validation data to estimate the likelihood of accurate reporting as a function of observables (e.g., by university, field, and/or year), and to weight our analysis with inverse propensity weights (IPW), overweighting low propensity populations to obtain representative results—similar to how survey weights are used to attempt to make non-representative survey results reflect the underlying population, or how research using nonrepresentative linked samples reweights on link propensities (Bailey et al. 2020). Across the paper, we opt for reporting the raw data to prioritize transparency and conservatism, and present IPW-weighted results in Appendix B.[19] Our results should thus be treated as a lower bound on underlying (true) rates of support.

Despite these limitations, one final point of comparison is available from Chang et al. (2019), who link SED graduates from seven Midwestern universities between 2011-2012 to UMETRICS data in order to evaluate the share of SED graduates at these universities who were paid with federal funding in the final two years of their PhD—including through research assistantships, teaching assistantships, and fellowships. Much of Chang et al. (2019)'s analysis presents data for all graduates (rather than STEM graduates), but in passing reports that "over three in five students in physical sciences, agriculture/biology, and computer/mathematical sciences are on federal grants in the two years prior to getting their degree" (Chang et al. 2019, p. 1490). By comparison, our IPW-weighted measures indicate 60% of STEM graduates from these university-years report federal support in their dissertations (65% of graduates in physical sciences, 62% in life sciences, and 41% in math and computer science)—consistent with the UMETRICS data.

4 New Facts about U.S. STEM PhD production

4.1 Who funds PhD training in the United States?

Figure 4, Panel (A) shows the share of graduates supported by different organization types from 1950 to 2022. In 2022, approximately 50% of graduates in our sample acknowledged some source of support. U.S. federal agencies are the most common funders, supporting over 40% of graduates. Roughly 15% of graduates are supported by non-profit organizations, and 10% by private firms. These patterns have changed substantially over time: government support increased rapidly in the 1950s and 1960s, peaked around 1968, retracted in the 1970s, and stabilized (at ≈40%) in 1980. It has since modestly grown (2000 onwards). Non-profit and firm support has also fluctuated; the

[19] In exploring the data descriptively, we have found that within our ground truth sample, variation in reporting cannot be explained by cross-institution differences (which might have been generated by, e.g., institutional dissertation templates or guidance), as most institutions have similarly middling reporting rates. Major field turns out to have the most explanatory power for reporting, and we base our estimated propensity off of these.

share of graduates supported by firms peaked at 20% around 1960, whereas nonprofit support grew from 10% to 15% between 1980 and 2000 and has since stabilized.

Panels (B1) to (B4) disaggregate these patterns by broad field. Disaggregation reveals that government support was historically especially high in the physical sciences and engineering, but has since largely equalized across most fields. Non-profit support is somewhat higher in the life sciences, and industry support in engineering, though these patterns have also fluctuated: firms, for example, supported 30% of graduates in the physical sciences in 1960—which follow-up analysis indicates was led by Du Pont in chemistry—but under 10% today; conversely, non-profits supported 10% of life sciences graduates in 1980—but more than 20% today.

[Figure 4 about here]

Figure 4 is almost certainly a lower bound on true rates of support, which are undermeasured due to underreporting. In Appendix Figure B.7 we show IPW-weighted shares (i.e., attempting to adjust for this underreporting), which we treat as an upper bound, and which indicate that in 2022, as many as 67% of graduates may have had U.S. government support, 22% non-profit support, and 16% industry support. Using the raw and IPW-adjusted shares, we plot confidence bands, along with the midpoint. Though IPW-weighted values are larger than raw values, the variation across domains and over time remains similar. The appendix also evaluates other dimensions of heterogeneity, including by country of origin, which we observe for a subset of our sample (inferred from pre-doctoral institutions, which are reported in many dissertations, especially post-2000); in Appendix Tables B.2 and B.3, we show that a substantial share of both U.S. and foreign-origin students have U.S. government support, though substantially more U.S.-origin students do—reflecting their eligibility for a broader range of federal support mechanisms.

Table 2 lists the top 15 sources of external support for U.S. PhD graduates, by sector (government, industry, non-profit). The National Science Foundation (NSF), National Institutes of Health (NIH), Department of Defense (DoD), and Department of Energy (DOE) rank high on this list, as do the National Aeronautics and Space Administration (NASA) and Department of Agriculture (USDA). Top non-profit funders include academic societies and philanthropic foundations. The top firms tend to be large, research-intensive multinational corporations. Intel and IBM top this list, but the most represented industries are the chemical, oil & gas, and pharmaceutical industries—indicating that many major corporations in high-tech manufacturing industries invest in developing highly-trained researchers who might augment the stock of relevant knowledge or might contribute directly to their future R&D efforts as future employee researchers.

[Table 2 about here]

4.1.1 Heterogeneity across subjects and universities

Figure 4 points to heterogeneity in the level of support broad subjects receive from different categories of funders. In Appendix B.4 we explore these differences in finer detail, plotting the share of PhD graduates between 2000 and 2022 in specific subjects with government and industry support. Appendix Figure B.5 splits the sample into very large subjects (with >1200 graduates in this period; Panel A) and large subjects (400-1200 graduates; Panel B) to improve readability. The figure provides insight into which areas of science are predominantly funded by either firms or the federal government. Scientific training in automotive engineering and pharmaceutical science, for example, is more heavily industry-supported than government supported. Astronomy and astrophysics, on the other hand, are heavily government-supported and receive essentially no private support. Geology and materials science receive a mix. In nearly all subjects, however, the federal government provides much more support (5 to 10x) of doctoral training than industry.

Appendix Figure B.6 provides similar analysis by university, plotting the share of STEM PhD graduates between 2000 and 2022 in each university with government and industry support. We similarly split the sample into very large universities (with >2000 graduates in this period; Panel A) and large universities (1000-2000 graduates; Panel B). The figure shows a clear positive correlation in rates of government and industry support—suggesting that the doctoral funding system concentrates among the same institutions and students. Technical universities like MIT, Carnegie Mellon, Georgia Tech, and Virginia Tech tend to have higher rates of industry support, but as before, the federal government remains (by far) the larger sponsor of PhD training.

4.1.2 Patterns in government support by gender and nationality

The granularity of our data support analysis of heterogeneity across other margins, including individual student characteristics. In Appendix B.1, we use information on pre-PhD education to identify probable foreign nationals and evaluate the rates at which they report U.S. government support. Although PQDT does not report nationality, many dissertations list where the author received lower degrees (often undergraduate, occasionally master's), seemingly on the basis of the dissertation template they are given at the time of filing. We find this information in approximately 10% of dissertations in our sample, and 18% of those post-2000, including at high frequency at dozens of institutions. Using lower degree institution locations as a proxy for origin country, Appendix Tables B.2 and B.3 examine the share of graduates in each major field and critical technology area who report a degree from a foreign pre-doctoral institution, and in turn the share of these (likely-foreign) graduates who report U.S. government support.[20]

[20] We are grateful to an anonymous reviewer for suggesting this measurement approach.

The (apparent) foreign-origin share of U.S. PhD graduates varies substantially across fields, with ~20% in the biological sciences having a non-U.S. BA (or equivalent), and ~60% in civil engineering—rates which are directionally similar to the SED. Whereas 50% of AI graduates and nearly 60% of data and cybersecurity graduates have a non-U.S. BA, only ~25% of those related to space technology or hypersonics appear foreign. Perhaps most interestingly, U.S. lower degree holders report government support at higher rates than foreign degree holders—in part due to eligibility restrictions for certain categories of federal support—but non-U.S. degree holders also report U.S. government support for their PhD research at high rates, reflecting that the U.S. government funds PhD research broadly and through a wide array of mechanisms.

4.1.3 Specialization and variation over time at federal agencies

We next turn attention to specific sources of support—especially federal government agencies, focusing on the four largest government supporters of PhD training: NSF, HHS, DoD, and DOE. Figure 5(A) characterizes the share of graduates in each of 17 major fields funded by these agencies. NSF provides relatively even support across subjects (except for health sciences, reflecting NSF's prioritization of funding basic science) and is an especially large supporter of students in geological (i.e., earth, atmospheric, and ocean) sciences. Other agencies have a clear mission focus. HHS, for example, supports a large fraction of trainees in biological and health science and biomedical engineering, but few in other fields. DoD is the country's largest funder of doctoral training in aerospace engineering and a large supporter of electrical engineering and computer science. DOE emphasizes physics, chemical engineering, and materials science. Though not shown, similar patterns are present for other agencies—for example, USDA is a major sponsor of agricultural sciences; DOT, of civil engineering; and NASA, like DoD, of aerospace engineering.

[Figure 5 about here]

Figure 5(B) shows how this support has varied over time. The common pattern across most agencies is that their support for PhD training peaked in the late 1960s and has only recently recovered, if at all. Roughly 20% of STEM PhD graduates currently acknowledge support from NSF, 15% from HHS, and 12% from DoD and DOE collectively (roughly 5-7% individually)—shares which we consider conservative (lower bounds, given the potential for underreporting, as discussed above), but which capture the relative rate of support across agencies.

4.2 PhD production in critical technology areas

An additional important dimension of the U.S. STEM scientific training ecosystem is the cultivation of scientific expertise in critical technology areas. Using the measures introduced in

Section 3 we tabulate the number of graduates in our sample associated with each of the 18 OSTP (2024) critical and emerging technology areas by university and source of support between 2000 and 2022. Table 3 lists the top five universities and sources of support for graduates in each of the 12 largest technology areas (excluding AI, which we will examine in more detail momentarily).[21] Despite the ostensible importance of critical technology science to U.S. science policy and economic policy, to our knowledge, for most of these technologies the U.S. doctoral education system has not been systematically and nationally assessed to understand how many such scientists the U.S. trains, where they are trained, and who pays for their training.

[Table 3 about here]

The table offers many insights. First, we find that several universities rank highly in multiple technologies areas, including MIT (in the top 5 for 10 of 12 areas in the table), Stanford (9), and UC Berkeley (7)—highlighting that these universities play an outsize role in the U.S. innovation ecosystem. Maryland, Michigan, Purdue, and UCLA also rank highly for 4+ technology areas. Second, we observe that some universities have strength in specific subjects, some of which are a product of decades of public investment. MIT, Northwestern, NC State, Penn State, and UIUC are the top universities for training PhDs in advanced materials—and all were among the 12 institutions which were selected by DARPA to establish materials science laboratories in the 1960s, when the field was effectively born. Third, with little exception, U.S. federal agencies are the top sources of support for trainees in all of these areas. For all of these facts, it is important to keep in mind that a few universities which are important training grounds for scientists in these areas (e.g., Georgia Tech) may be omitted from the list due to their exiting the PQDT sample early (as seen in Appendix Table A.12). Appendix Table B.6 reproduces Table 3 with data through 2012 only to explore what some of these may be, but the comparison also elucidates how the relative importance of each university to these technology areas may have changed over time.

Table 4 provides a similar analysis for artificial intelligence specifically, expanding the list of reported universities and research sponsors. At the top of the list of universities training researchers in AI are many expected institutions (e.g., Stanford, MIT, and UC Berkeley), which further reinforces the face validity of the method. Also in the top 15 schools, however—and especially below the top 5—are many public universities, which collectively train more PhDs in AI than the top 5 schools. As before, a small number of schools which might have otherwise made this list (e.g., Georgia Tech) are likely absent because they exited the PQDT sample early—an intuition supported by

[21] Technology areas omitted from this table are: human-machine interfaces, positioning, navigation, and timing (PNT) technologies, advanced gas turbine engines, directed energy, and hypersonics.

Appendix Table B.7, which reproduces this analysis through 2012 (despite being prior to the recent AI renaissance) and finds Georgia Tech ranked in the top three.

[Table 4 about here]

The right half of Table 4 shows that the top six sources of support for AI PhD trainees are government agencies, with NSF and DoD atop the list. The next-largest source of support is large U.S. technology companies such as Google, Microsoft, IBM, Intel, and Nvidia. The differences in magnitude between government and private support, however, are very large: the NSF supports 10x as many graduates as Google, and 20x as many as Intel.

In Figure 6, we aggregate this information up to the technology level, documenting the share of graduates in each technology area reporting government support against the total number of graduates in that area between 2000 and 2022. We color code each technology by the principal agency providing support. Figure 6 shows that there are essentially three clusters of PhD graduates with respect to critical technologies: the defense cluster, relatively low in count but heavily DoD-supported; the civilian mission agency cluster (space, energy, and biotech); and the NSF cluster, which spans a range of subjects but is concentrated around advanced electronics, communications, and computing. NSF is the leading patron of two-thirds of the areas in the OSTP list, and two-thirds of the graduates we associate to any critical technology area.

[Figure 6 about here]

4.3 Effect of government funding on PhD production

The very large role that federal funding plays in supporting scientific training in STEM, including in critical technology areas—together with historical fluctuations in the level of public support, including recent cuts (at the time of writing; Bhatia et al. 2025)—naturally raises the question of what happens to PhD production when federal research funding programs expand or contract. Does the production of new scientists change with it? If so, how much?

To study these questions, we use our data to count PhD graduates at the field level and relate annual PhD production to the level of federal support. We estimate the following relationship, where i indexes fields and t indexes years; $PhDGraduates_{it}$ is total graduates in field-year it; $SupportedGraduates_{it}$ is observed graduates in field-year it with federal support; X_{it} represents additional controls; and α_i and δ_t are field and year fixed effects, with a baseline estimation sample spanning all 17 STEM major fields and the years 1970 to 2022:

$$\ln PhDGraduates_{it} = \beta \cdot \ln SupportedGraduates_{it} + X_{it}\phi + \alpha_i + \delta_t + \varepsilon_{it}. \tag{1}$$

We also estimate a variant of this specification at the university-field-year level, where our panel units become university-fields, which affords added precision.

The focal parameter throughout this analysis is β, which represents the elasticity of total PhD production with respect to government-supported graduates in a given cohort. The value of β will indicate whether increases in government-supported graduates displace other graduates (crowd-out), do not affect the number of other graduates (additivity), or increase other graduates (crowd-in). However, this interpretation cannot be made from the estimate of β directly, but rather is a function of the baseline government-supported share. To see this, let T be shorthand for total graduates, G for government-supported graduates, and $N(G)$ for others, where $T = G + N(G)$. The crowd-in question is one of the derivative $N'(G)$, which could be positive, negative, or zero, or equivalently whether $\frac{dT}{dG}$ $(= 1 + N'(G))$ is greater than, less than, or equal to one.

Observe that β in Equation (1) can be written as $\beta = \frac{d\ln T}{d\ln G} = \frac{dT}{dG}\frac{G}{T}$, such that $\frac{dT}{dG} = \frac{\beta}{s}$, where $s \equiv \frac{G}{T}$. With estimates of β and s, $\frac{dT}{dG}$ is thus implied. In our data, s appears to be between 0.4 (as observed; Figure 4) and 0.67 (adjusted for potential underreporting; Appendix Figure B.7), which will bracket the potential crowd-in for any estimate $\hat{\beta}$.[22]

4.3.1 Identification and shift-share design

Evaluating the relationship between investments in PhD training and PhD production presents two challenges. The first is the threat of a mechanical relationship in a specification where the outcome is total graduates and the explanatory variable is a category of graduates. The second is that this relationship may be confounded by unobserved factors. For example, federal support may flow to fields with rising scientific opportunity, which may also attract students regardless of the level of government support and lead to overstating the effect of federal funding on PhD production. Conversely, federal agencies may direct support toward mission-relevant fields precisely when those fields are undersubscribed, which would lead us to understate it.

To confront these challenges, we use a shift-share design to instrument for supported graduates

[22] Underreporting will generally not affect estimates of β, as it effectively acts as a scaling parameter between actual vs. observed $SupportedGraduates_{it}$ and will be additively separated by the log transformation and absorbed into fixed effects. Underreporting instead affects how we convert $\hat{\beta}$ into an estimate of crowd-in, per the arithmetic above. Note that we can also evaluate crowding by estimating Equation (1) with non-government supported graduates as the outcome, but doing so would arrive at the same findings, due to the accounting identity $T'(G) = 1 + N'(G)$. We estimate Equation (1) as written because total graduates is our primary object of interest, and because this allows us to use our measured government-supported share s to compute implied crowd-in.

(e.g., Goldsmith-Pinkham et al. 2020, Borusyak et al. 2022), combining each agency's annual number of supported graduates (the shift) with each field's share of that agency's graduates in a predetermined base period (the share). This design exploits the variation in Figures 5(A) and 5(B)—the combination of which creates variation in federal support in each field over time. For example, changes in total DoD and NIH support will (ostensibly) have different effects on aerospace engineering versus biology, given their respective specializations.

Concretely, we (i) calculate field i's share of supported graduates from each agency a in a base period (ω_{ia}); (ii) interact it with the log number of agency a's supported graduates in t, computed over all fields other than i ($Graduates_{at}^{-i}$, a leave-one-out shifter); and (iii) sum across agencies to get field i's predicted log number of supported graduates in year t:

$$\ln \widehat{SupportedGraduates}_{it} = \sum_{a} \omega_{ia} \cdot \ln Graduates_{at}^{-i} \tag{2}$$

We define the base-period shares ω_{ia} in three ways, each predetermined relative to the cohorts they instrument: averaging over (i) the 6 to 10 years prior to year t (rolling), (ii) 11 to 15 years prior (rolling, granting more distance), and (iii) a static 1960-1969 window. In the static case we restrict the estimation sample to post-1980, ensuring at least a decade between the base period and the earliest cohort. We condense the agency set to the six largest federal funders (DoD, DOE, HHS, NASA, NSF, and USDA), an "other agencies" category, and a "multiple agencies" category to avoid double-counting graduates with more than one federal sponsor. In university-field-year analysis, we construct an analogous variant of this instrument at this level.

Even with this instrumental variable design, residual identification threats can remain, especially if agency budgets fluctuate based on the evolving potential of the fields they support—such as if Congress directs resources to agencies that specialize in a given field when that field's potential is elevated. Two features of our analysis are designed to reduce this risk. The first is our use of leave-one-out shifters, which avoid direct contamination from same-field changes; the evolving opportunity in field i would then have to be correlated with that of other fields in the same supporting agencies' portfolios to confound. The second is that our implementation of Equation (1) will include field × year fixed effects (broad field × year in the field-year analysis, narrow field × year in the university-field-year analysis), which will absorb evolving field-specific factors that might correlate with the instrument and independently drive PhD enrollments.

Though not fully complete—as opportunities may shift over time at levels below our measurement resolution (subfields)—the leave-one-out shifter together with these fixed effects do substantial work to account for latent threats which the shift-share design would not account for. Additional robustness checks in Appendix B.7 will further reinforce that the results are not driven by any particular field (or its subfields). More broadly, agency-specific fluctuations have also often been driven by exogenous factors that shaped the resources they receive (e.g., reductions in defense R&D funding after the end of the Cold War (post-1991), stimulus funds from the American Recovery and Reinvestment Act (2009), or the politics of biomedical research).

4.3.2 Estimation results

Table 5 presents OLS and 2SLS estimates of Equation (1) under each instrument, at the field-year level (Columns 1-4) and the university-field-year level (Columns 5-8). The estimated elasticity of total PhD production with respect to federally-supported graduates is close to one and broadly consistent across columns: estimates range from roughly 0.8 to 1.1 under the 2SLS specifications, with first-stage F-statistics well above conventional thresholds. Our preferred specification is the university-field-year level estimate in Column (8), generated from the 1960-1969 base year instrument, which we consider our most econometrically conservative result.

[Table 5 about here]

At an observed (overall) government-supported share of ∼40%, $\frac{dT}{dG} \approx 2$, implying that for every government-funded graduate, there is an additional unfunded graduate. At a reweighted (for potential underreporting) share of ∼65%, $\frac{dT}{dG} \approx 1.2$, implying that for every five government-funded graduates, there is an additional unfunded graduate. Our interpretation is thus that the evidence is inconsistent with crowd-out, and instead consistent with additivity or modest crowding in—though given the range above we cannot confidently claim a precise value.[23]

Appendix B.7 presents a range of additional robustness checks. We first decompose our instrument into a scale and an allocative component, where the scale component represents a fields' differential exposure to swings in total government support and the allocative component represents fields' exposure to changes in the cross-agency composition of the federal portfolio; for example, physics has a large scale factor loading, as it is supported at relatively high rates by many agencies (DoD, DOE, NASA, NSF), whereas biomedicine has a large allocative factor loading, being sensitive to changes in NIH's share of total government support. We find that the allocative factor explains our variation on its own, and controlling for scale (i.e., for individual field's scale of federal support), the results are unchanged. We then perform additional tests to push on allocative confounds (e.g.,

[23]In theory, crowd-in can operate through expansions in capacity and returns to scale—for example, if (i) government-funded graduates are supported under grants that fund faculty, equipment, or infrastructure that other students also use or train from, (ii) grants fund overhead that is spread across programs and students, or (iii) government funding encourages private (philanthropic or industry) funding, whose marginal productivity grows because it can leverage assets that federal funding engenders (e.g., Azoulay et al. 2026).

if NIH support grows specifically when biomedical research possibilities flourish). First, presuming that residual opportunity confounds would concentrate in the fields closest to recent technology booms, we re-estimate the relationship excluding four such fields jointly (computer science, electrical engineering, biomedical engineering, and the biological sciences). Second, we re-estimate our focal specification dropping one field at a time, for all fields. The estimates are essentially unchanged—indicating that no single field, nor a group of technologically proximate fields, drives the result. Finally, we show that the results remain quantitatively and statistically similar when estimated on a sample restricted to universities present in PQDT throughout the estimation period, confirming that the results cannot be attributed to any sample attrition.

These results should nevertheless be interpreted cautiously, given that instrumental-variable designs rely on hard-to-test assumptions of conditional independence. Our view is that the evidence is suggestive rather than dispositive: $\hat{\beta}$ statistically sustains under multiple instruments, withstands demanding controls, and does not depend on any single field, but it rests on instrument shares being independent of unobserved field-specific shocks that would not otherwise be absorbed by fixed effects. Additionally, given the relatively aggregated level at which this analysis is conducted, the estimated elasticity represents a bundled, equilibrium response of the training ecosystem to public investment—one that includes the induced expansion of non-federally supported graduates implied by the crowd-in discussed above—rather than a direct impact.

Comparisons to existing literature

The evidence is broadly consistent with prior research finding that fellowship applications and PhD enrollment respond to the availability of funding (e.g., Freeman et al. 2009, Blume-Kohout and Adhikari 2016). The closest prior analysis is arguably Blume-Kohout and Adhikari, who relate NIH-supported PhD students to total PhD enrollment in biomedical sciences after the early 2000s doubling of the NIH using NSF GSS data and find that "NIH-funded traineeships and fellowships increase full-time graduate enrollment by essentially 1:1," with more attenuated effects for research assistantships (Blume-Kohout and Adhikari 2016, p. 1297). Our finding that total PhD production rises roughly in proportion to federally-supported graduates, with no evidence that other graduates are crowded out, is consistent with these findings, while adding a more systemic, population-level perspective spanning multiple federal funders and long panels.

We view our estimates as adding an additional datapoint on the relationship of public funding to national PhD production—and on the impacts of public R&D more broadly—rather than a singularly decisive result. Differences in samples, measures, sources of variation, and specifications can all generate differences in point estimates, potentially reflecting meaningful heterogeneity. Moreover, as others have shown (e.g., Kim et al. 2022, Graddy-Reed et al. 2021), these elasticities can vary by funding mechanism, and funding can affect not only enrollments but completion, placement, future productivity, and academic networks. Because our measures speak to overall levels of support rather than specific mechanisms through which students were funded or otherwise supported, we abstract from this heterogeneity to emphasize the total size and breadth of the PhD graduate population as a broad indicator of evolving scientific capacity (OECD 2015).

5 Distribution Dataset

Using the above-described data, we compile three public-use datasets measuring PhD production by university, field, and university-field over time. In all three cases we limit the sample to universities which have ever held an R1 (very high research activity) or R2 (high research activity) classification post-2000, or which are dedicated medical schools (e.g., SUNY Downstate or UT Southwestern). For each dataset, we report the total number of PhD graduates in the natural sciences and engineering; the number for which we have full text and for which we have detected acknowledgments (for normalization); the number we identify as supported by the U.S. government, by industry, or by non-profit organizations; and the number supported by specific federal agencies (DoD, DOE, HHS, etc.). For the university-level dataset we further report the number of graduates we associate to OSTP critical technology areas. Data and accompanying documentation have been posted to the Harvard Dataverse and are available for public use (Shvadron et al. 2025a).

The distribution data include information for 338 unique institutions between 1950 and 2022—though the sample is unbalanced, with roughly 100 institutions in the 1950s and nearly 300 by the end of the sample. The field and university-field datasets report graduates for the 17 focal SED major fields described earlier, spanning the life sciences, physical sciences, and engineering. Users of these data who wish to make use of the counts of students with particular sources of funding may want to normalize these counts by either (i) the number of graduates in the given university-year, field-year, or university-field-year for which we have dissertation text (variable `has_ft`), or (ii) the number for which we have identified acknowledgments (`has_ack`).

6 Conclusion

In this paper, we use information from the dissertations of nearly 1.2 million U.S. STEM PhD graduates between 1950 and 2022 to examine the U.S. scientific training ecosystem. Leveraging recent advances in LLMs, we (i) link each dissertation to 18 critical and emerging technology areas recently identified as national priorities by the White House OSTP and (ii) process dissertation acknowledgments to identify research sponsors. With these data we show that about half of recent graduates acknowledge external support, with the U.S. federal government supporting about 42% of graduates—far eclipsing industry (≈10%) and non-profits (≈15%). NSF and NIH alone individually account for more supported PhDs than the entire commercial sector, and their support is highly concentrated in universities that also dominate critical-technology training (e.g., MIT, Stanford, UC Berkeley, among others). Across the 18 OSTP domains, federal agencies are typically the top funders, though a range of universities train students in these fields.

Exploiting agency-specific variation (in both the primary fields supported and total number of graduates supported over time) in a shift-share design, we estimate an elasticity between government support and PhD production near one, indicating that total PhD production scales proportionally with the number of government-supported graduates, and with a magnitude that implies additivity or modest crowd-in—underscoring public investment as a driver of expansions and contractions in the scientific workforce. These data, methods, and findings together provide a new, scientifically- and policy-relevant map of the U.S. scientific training ecosystem and a foundation for tracking how funding choices influence national science and innovation capacity. The approach we develop can also be extended forward and to other national innovation systems.

Whether more or less public investment (and accordingly, more or fewer PhD trainees) is desirable is not a question this paper specifically speaks to: our goal is to document features and effects. In particular, as in other markets, public subsidies may distort market outcomes. Nelson (1959), Arrow (1962), and others have argued that public investment in R&D can correct inefficiencies more than it introduces them, and policy advisors as far back as Bush (1945) have made the case for funding scientific training. Yet a perennial question and point of concern is whether the U.S. innovation system trains too many PhD scientists. Critics point to the limited number of tenure-track faculty jobs relative to PhD graduates, large number of postdocs, length of postdocs, and the growth of contingent faculty as evidence of oversaturation (e.g., Cyranoski et al. 2011, Powell 2015), especially in light of the prevailing wages these workers accept (Stephan 2013).

On the other hand, the size of the scientific workforce is typically seen as the rate-limiting factor for scientific progress, and in turn for improvements in economic growth and material well-being (Steelman 1947, Atkinson 1990, Romer 1990, Jones 2009). Those who advocate growing the scientific workforce point out that many PhDs enter industry, where they can be important contributors to private sector innovation (Scharfmann et al. 2025). Firms are generally considered unlikely to invest in transferable skills like scientific training, since those skills are broadly marketable (Becker 1964). This intuition may explain why industry funds relatively few PhDs (Table 2)—and where it does, it often only supports science with specific private value (as our example in Figure 3 demonstrates). Given this logic, and evidence that the returns to public R&D (much of which funds training; Sattari et al. 2022) are often found to be very high (Jones and Summers 2022, Fieldhouse and Mertens

2024), it is easy to make the case for growing the scientific workforce.

We do not stake a claim on the issue, but rather observe that the answer may hinge on which scientists we have in mind—and the data we produce can be helpful to assessing seemingly important categories of trainees. Moreover, our evidence reinforces that public funding plays a central role in determining their supply (Table 5; also cf. Dugoua et al. 2025), and that the data we introduce can be used to further adjudicate this debate—including the possibility that whether public subsidy corrects or distorts markets may vary over time and by field, with critical and emerging scientific fields being riper targets for public investment than more mature ones.

The methods and data of this paper point to other opportunities for future research. The methods we develop can be used more widely to evaluate the content of dissertation science or other scientific corpora directly from the text. Our entity recognition procedure can also be applied more broadly, and in ongoing research (Gross et al. 2025), members of this author team are adapting this approach to extract research funders from historical publications.

Beyond methods, the data accompanying this paper can be used in many ways, including for evaluating where scientific training takes place, how it has evolved over time, what drives changes in its scale and composition, and what its effects are on institutions, regions, scientific fields, and more. For example, how does PhD production interact with regional economies? To what degree do public investments in scientific training spill over into (i) closely related local industries, vs. (ii) less-related industries in the same region, vs. (iii) other regions? How important are PhD programs to universities' overall research productivity—and how much does research suffer when PhD enrollment declines or programs end? How important are PhD programs to nearby firms—and conversely, what is the role of industry support in doctoral training? Do universities with more industry-funded PhDs produce more commercially oriented research, industry placements, or university startups? And backing out from these more localized effects: what is the importance of PhD production to the wider U.S. innovation system, and what happens to U.S. science and innovation as the PhD population ebbs and flows? In releasing data which maps the U.S. scientific training ecosystem over a long horizon, we hope to enable further research on these questions.

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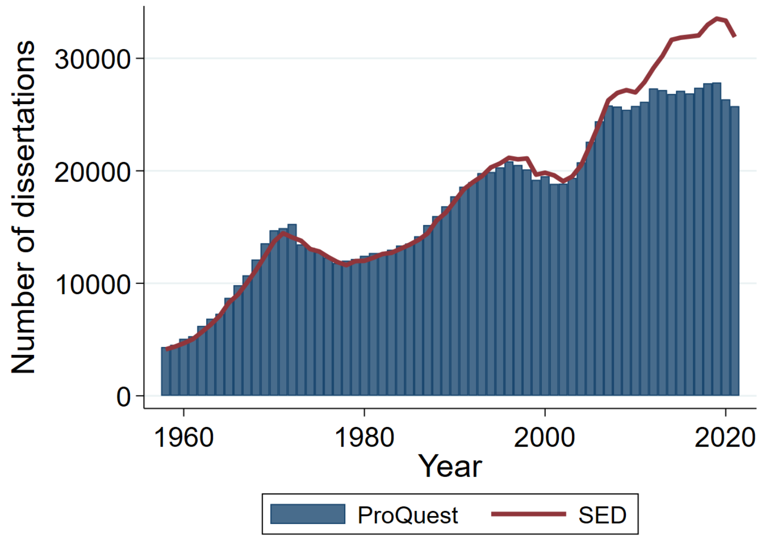
Figure 1: ProQuest vs. SED graduates in STEM fields, by year

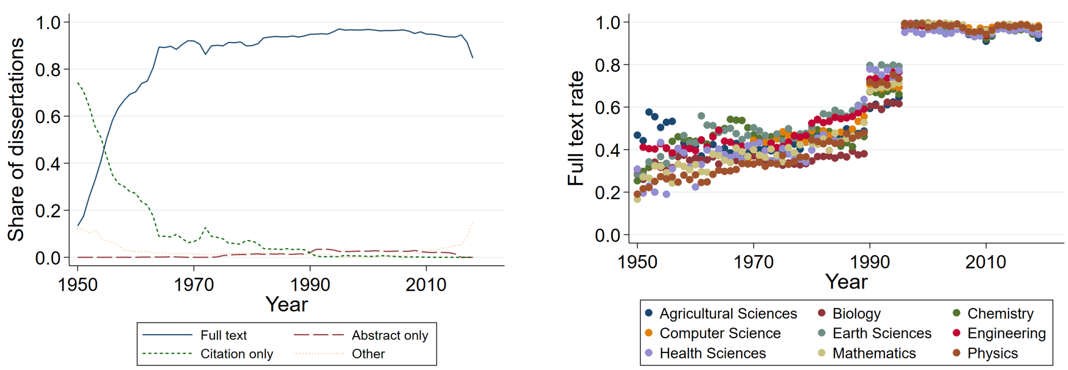


Notes: Figure shows annual PQDT graduate counts in the physical and life sciences and engineering (blue bars) and SED counts (red line) for comparison, from 1958 (the first year of the SED) to 2022.

Figure 2: Availability of dissertation text

Panel (A): Annual share of graduates in each dissertation indexing category

Panel (B): Full text availability in practice, over time and across major fields

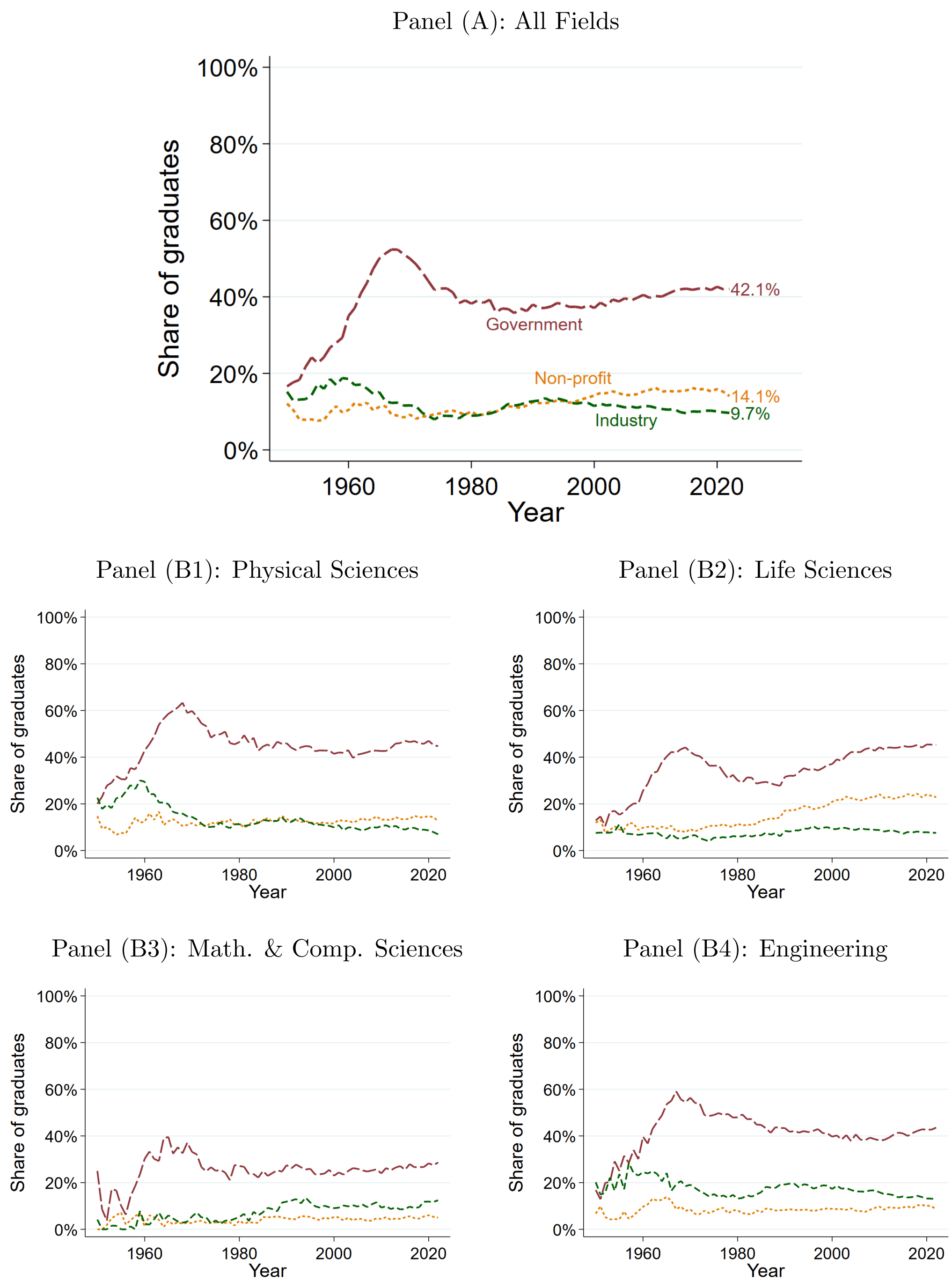


Notes: Panel (A) shows the annual share of dissertations in our sample that ProQuest identifies as having been indexed full text (i.e., ProQuest received metadata (author, title, subject, etc.) and a full copy of the dissertation), abstract-only (metadata plus abstract), or citation only (metadata only). Panel (B) shows the annual share for which we actually have full text.

Figure 3: Example dissertation acknowledgment (Ian Buck, Stanford University, 2005)

I have had a variety of funding sources throughout my graduate career including DARPA, DOE, ASC, NVIDIA, ATI. Specifically, I would like to thank Randy Frank, Bob Graybill, and Mike Macedonia for putting money where my mouth was. I also have been fortunate in receiving fellowships from the Stanford School of Engineering as well as NVIDIA.

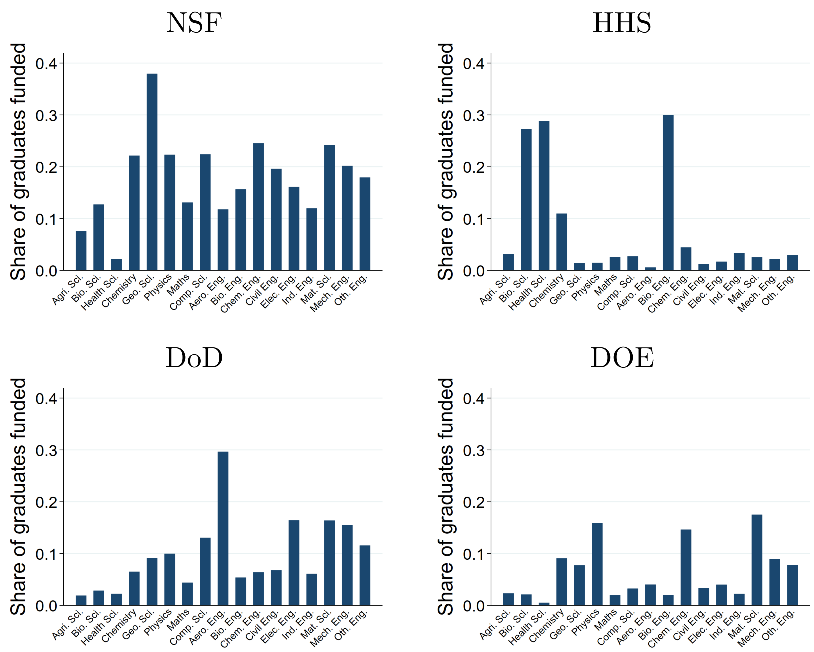
Figure 4: Share of PhD graduates supported over time, 1950-2022, by organization type, based on acknowledgments reported in dissertations (a lower bound)



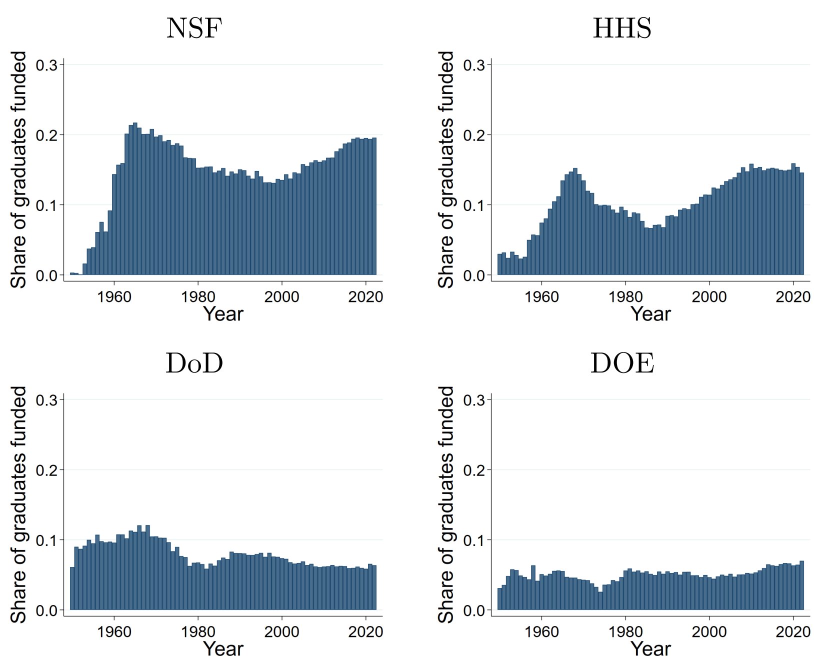
Notes: Figure shows the share of PhD graduates reporting their sources of support who are supported by (i) U.S. federal government agencies, (ii) firms, and (iii) non-profit organizations, from 1950 to 2022. Panel (A) presents overall results. Panels (B1) to (B4) present results by broad field (physical sciences, life sciences, mathematical and computer sciences, and engineering). Sample restricted to dissertations with acknowledgments (>98% of all dissertations). The shares shown here should be interpreted as a lower bound due to underreporting. See text for discussion, and see Appendix Figure B.7 for upper bounds, which use inverse propensity weighting to account for underreporting.

Figure 5: Federal agency variation in PhD graduate support, across fields and over time

Panel (A): Major fields' share of graduates supported by select federal agencies, 1950-2022

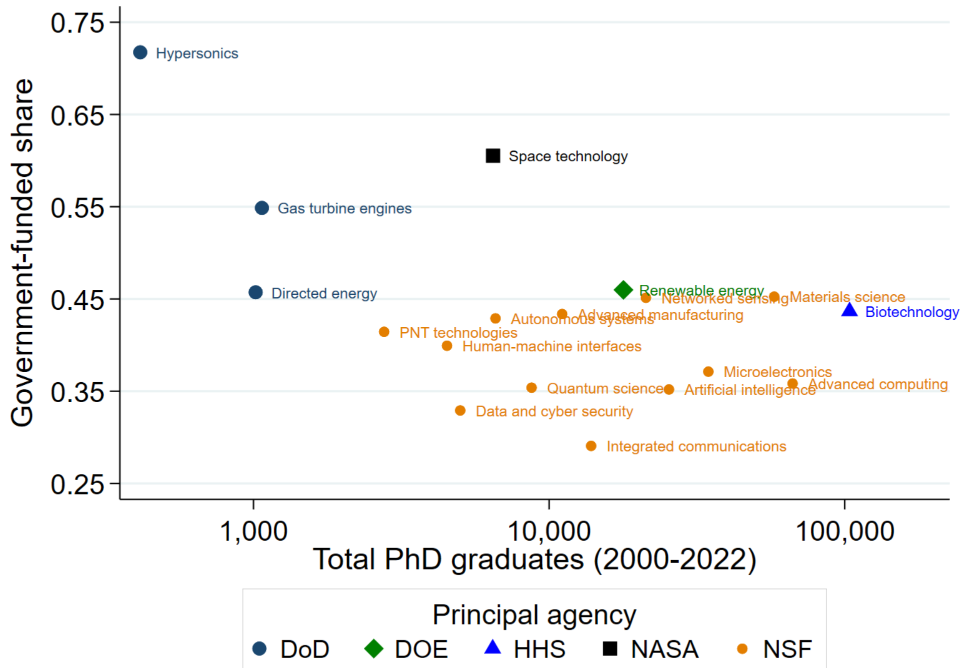


Panel (B): Share of all graduates supported by select federal agencies over time, 1950-2022



Notes: Panel (A) shows the share of PhD graduates in individual major fields supported by each agency listed (NSF, HHS, DoD, DOE), illustrating differences in subject matter priorities across agencies. Panel (B) shows the share of PhD graduates supported over time by each agency listed (NSF, HHS, DoD, DOE), illustrating long-run changes in these agencies' investments in PhD training.

Figure 6: Share of graduates reporting government support, by critical technology area, 2000-2022



Notes: Figure plots the share of graduates in each critical technology area reporting government support against the number of graduates in that technology area, color-coding each technology area's principal source of government support (DoD, DOE, HHS, NASA, or NSF). See OSTP (2024) for a complete list.

Table 1: Example extraction of acknowledged organizations for Ian Buck (Figure 3)

Entity Name	Entity Type	Support Type	ROR Organization
DARPA	government	funding	Defense Advanced Research Projects Agency
DOE	federal agency	funding	United States Department of Energy
ASC	unknown	funding	–
NVIDIA	private company	funding	Nvidia
ATI	private company	funding	Advanced Micro Devices
Stanford School of Engineering	academia	fellowship	–
NVIDIA	private company	fellowship	Nvidia

Notes: Table lists organizations identified in the text in Figure 3 as dissertation sponsors using our LLM-based procedure and provides their links to ROR. The LLM associates ATI with Advanced Micro Devices (AMD) because ATI Technologies was acquired by AMD in 2006.

Table 2: Top 15 acknowledged organizations, by sector, 2000-2022

Government agencies		Firms		Non-profit organizations	
Name		Name	Count	Name	Count
National Science Foundation	91,813	Intel	2,276	Howard Hughes Medical Institute	3,731
Department of Health and Human Services	77,989	IBM	1,942	American Heart Association	3,676
Department of Defense	34,081	Merck	1,417	Sigma Xi	2,787
Department of Energy	30,528	Google	1,300	American Cancer Society	1,568
National Aeronautics and Space Administration	15,038	Microsoft	1,221	American Chemical Society	1,453
Department of Agriculture	13,443	Pfizer	1,177	Geological Society of America	1,386
Department of Commerce	8,925	General Electric	977	Robert Wood Johnson Foundation	1,178
Department of the Interior	6,887	Du Pont	873	W. M. Keck Foundation	1,104
Environmental Protection Agency	5,630	Dow Chemical	847	Fulbright Program	953
Department of Transportation	5,283	Eli Lilly	822	David and Lucile Packard Foundation	943
Department of Education	4,615	Chevron	820	Welch Foundation	910
Department of State	4,244	ExxonMobil	774	Burroughs Wellcome Fund	897
Department of Veterans Affairs	2,218	GlaxoSmithKline	764	Gordon and Betty Moore Foundation	856
Agency for International Development	1,615	Novartis	753	Ford Foundation	815
Department of Homeland Security	1,368	Boeing	718	National Geographic Society	759

Notes: Table lists the top 15 government, industry, and non-profit sponsors of graduates between 2000 and 2022, as measured by our LLM-based procedure. Government and other sponsors acknowledged in dissertations are consolidated into ultimate parent agencies (e.g., defense agencies are grouped up to the U.S. Department of Defense (DoD); the National Cancer Institute is grouped into the National Institutes of Health, which is in turn grouped up to the U.S. Department of Health and Human Services, etc.).

Table 3: Top 5 universities and sponsors of PhD graduates in U.S. critical technology areas, 2000-2022

Advanced Computing

University	Count	Name	Sector	Count
Stanford U.	1,971	NSF	Govt	13,571
Mass. Inst. of Tech.	1,818	DoD	Govt	7,038
Purdue U., West Lafayette	1,617	HHS	Govt	4,400
U. California, Berkeley	1,572	DOE	Govt	3,411
U. California, Los Angeles	1,500	NASA	Govt	1,975

Advanced Manufacturing

University	Count	Name	Sector	Count
Mass. Inst. of Tech.	381	NSF	Govt	2,842
U. California, Berkeley	362	DoD	Govt	1,567
Purdue U., West Lafayette	326	DOE	Govt	1,064
U. Michigan, Ann Arbor	311	HHS	Govt	442
U. Illinois, Urbana-Champaign	296	NASA	Govt	341

Advanced Materials

University	Count	Name	Sector	Count
Mass. Inst. of Tech.	1,555	NSF	Govt	14,770
Northwestern U.	1,425	DOE	Govt	8,777
Pennsylvania State U.	1,345	DoD	Govt	7,714
North Carolina State U.	1,305	HHS	Govt	2,324
U. Illinois, Urbana-Champaign	1,290	NASA	Govt	1,666

Autonomous Systems

University	Count	Name	Sector	Count
Mass. Inst. of Tech.	327	NSF	Govt	1,403
Stanford U.	276	DoD	Govt	1,294
Carnegie Mellon U.	237	NASA	Govt	526
U. California, Berkeley	226	DOT	Govt	184
U. Michigan, Ann Arbor	203	HHS	Govt	166

Biotechnology

University	Count	Name	Sector	Count
U. Wisconsin-Madison	2,427	HHS	Govt	27,586
Harvard U.	2,308	NSF	Govt	18,226
Stanford U.	2,260	DoD	Govt	4,394
U. Washington, Seattle	2,082	DOE	Govt	4,091
U. California, Berkeley	2,006	USDA	Govt	4,007

Clean Energy Generation and Storage

University	Count	Name	Sector	Count
Stanford U.	528	DOE	Govt	4,541
U. California, Berkeley	519	NSF	Govt	4,031
Mass. Inst. of Tech.	506	DoD	Govt	1,343
Purdue U., West Lafayette	421	USDA	Govt	421
Pennsylvania State U.	415	NASA	Govt	400

Communications and Networking

University	Count	Name	Sector	Count
U. California, Los Angeles	500	NSF	Govt	2,512
Stanford U.	418	DoD	Govt	1,692
U. California, San Diego	384	Intel	Firm	322
Ga. Inst. of Tech.	361	DOE	Govt	314
Purdue U., West Lafayette	351	NASA	Govt	284

Data Privacy and Cybersecurity

University	Count	Name	Sector	Count
Purdue U., West Lafayette	164	NSF	Govt	1,189
U. Maryland, College Park	124	DoD	Govt	621
U. California, Los Angeles	101	DOE	Govt	143
Arizona State U.	95	Google	Firm	122
Mass. Inst. of Tech.	95	Intel	Firm	109

Microelectronics and Semiconductors

University	Count	Name	Sector	Count
Stanford U.	1,313	NSF	Govt	7,680
U. California, Berkeley	1,157	DoD	Govt	4,661
Mass. Inst. of Tech.	1,014	DOE	Govt	3,698
U. Illinois, Urbana-Champaign	857	Intel	Firm	904
U. California, Los Angeles	848	NASA	Govt	754

Networked Sensing

University	Count	Name	Sector	Count
Stanford U.	603	NSF	Govt	4,494
Purdue U., West Lafayette	572	DoD	Govt	3,173
Mass. Inst. of Tech.	527	NASA	Govt	2,147
U. Maryland, College Park	474	DOE	Govt	1,228
U. Michigan, Ann Arbor	472	HHS	Govt	725

Quantum Science

University	Count	Name	Sector	Count
Mass. Inst. of Tech.	382	NSF	Govt	1,985
Stanford U.	329	DoD	Govt	1,044
U. California, Berkeley	288	DOE	Govt	1,017
Harvard U.	277	HHS	Govt	164
U. Maryland, College Park	258	NASA	Govt	155

Space Technology

University	Count	Name	Sector	Count
U. Colorado Boulder	319	NASA	Govt	2,745
Stanford U.	221	NSF	Govt	1,265
Mass. Inst. of Tech.	220	DoD	Govt	836
U. Michigan, Ann Arbor	208	DOT	Govt	567
U. Maryland, College Park	201	DOE	Govt	374

Notes: Table lists the top 5 universities and sponsors of graduates between 2000 and 2022 in the 12 largest OSTP critical technology areas. Graduates allocated to technology areas as described in text. For a small number of schools we have limited data from the mid-2010s onwards (see Appendix Table A.12). These schools are thus undercounted, and with more complete data a few of these schools might have appeared more frequently in this table. In Appendix B we regenerate this table using data through 2012 only, which truncates the sample early but avoids systematically undercounting these schools. The results are broadly similar, though they highlight Georgia Tech as a university which ranks highly in many of these technology areas but is missing in this table from most of them, likely due to sample truncation (Georgia Tech stops reporting to PQDT after 2012).

Table 4: Top 15 universities and sponsors of PhD graduates in AI, 2000-2022

Top universities		Top funders		
University	Count	Name	Sector	Count
Stanford U.	760	NSF	Govt	4,925
Mass. Inst. of Tech.	708	DoD	Govt	2,758
U. California, Berkeley	611	HHS	Govt	1,920
U. Maryland, College Park	578	DOE	Govt	963
Purdue U., West Lafayette	564	NASA	Govt	651
Carnegie Mellon U.	562	Google	Firm	484
U. Washington, Seattle	535	Microsoft	Firm	327
U. California, Los Angeles	533	DOC	Govt	287
U. Michigan, Ann Arbor	476	DOT	Govt	283
U. Illinois, Urbana-Champaign	443	IBM	Firm	265
U. California, San Diego	435	Intel	Firm	252
Arizona State U.	417	Nvidia	Firm	228
U. Southern California	404	USDA	Govt	216
North Carolina State U.	370	Amazon	Firm	209
U. Minnesota, Twin Cities	368	Meta	Firm	162

Notes: Table lists the top 15 universities and sponsors of graduates between 2000 and 2022 in artificial intelligence (AI). Graduates identified as AI-related based on critical technology assessment (see Section 3).

Table 5: Relationship of PhD production to government-supported graduates, 1970-2022

	Field-year				University-field-year			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	OLS	2SLS	2SLS	2SLS	OLS	2SLS	2SLS	2SLS
Ln(USG-supported PhDs)	0.839***	0.982***	1.130***	0.993***	0.703***	1.118***	1.156***	0.781***
	(0.023)	(0.036)	(0.097)	(0.070)	(0.009)	(0.028)	(0.045)	(0.016)
N	901	901	901	731	70423	70423	70423	63954
F-stat		62.20	27.24	38.00		99.98	75.75	120.08
Field FEs	Y	Y	Y	Y				
Univ-Field FEs					Y	Y	Y	Y
Year FEs	Y	Y	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y	Y	Y
Sample	1970+	1970+	1970+	1980+	1970+	1970+	1970+	1980+
Instrument		Roll/6-10	Roll/11-15	1960-69		Roll/6-10	Roll/11-15	1960-69

Notes: Table estimates the relationship of annual PhD production by university, field, and year to government-supported PhD graduates. Columns (1) to (4) do so at the field-year level, and Columns (5) to (8) at the university-field-year level. Columns (1) and (5) estimate Equation (1) by OLS, and the remaining columns by 2SLS, using three shift-share instruments for government-supported PhD graduates, based on (i) agency field shares calculated from 6-10 years prior to a given year (Roll/6-10), (ii) shares calculated from 11-15 years prior (Roll/11-15), and (iii) a static, 1960-1969 base period (1960-69). Estimation sample covers the post-1970 period except in Columns (4) and (8), which are restricted to post-1980 to ensure at least a decade between the shift-share base period and the earliest cohorts in the estimation sample. Columns (1) to (4) additionally control for broad field × year fixed effects, and Columns (5) to (8) narrow field × year fixed effects, to absorb national trends in PhD production by field. First-stage F-statistics reported for the 2SLS columns. *, **, *** represent significance at the 0.1, 0.05, and 0.01 levels, respectively. Heteroskedasticity-robust SEs in parentheses in Columns (1) to (4) and clustered by university in Columns (5) to (8).